

Advanced Distributed Systems

Embodied Intelligence Systems Based on Individual and Group
Collaboration

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Figure III-1. Hybrid Distributed Multi-Agent System Architecture 错误！未定义书签。

I.Introduction

1.1 Research Background and Motivation

With the rapid advancement of artificial intelligence technology, we are at a critical turning point where it is deeply permeating both the "digital world" and the "physical world." Early AI research primarily focused on perceptual and cognitive tasks such as data understanding, image recognition, and natural language processing, with applications largely confined to virtual environments.[1][3] However, in recent years, endowing agents with the ability to perceive, understand, make decisions, and interact in real time with the physical environment — known as "embodied intelligence" — has become a shared focus in both academia and industry. The core of this capability lies in the agent no longer being merely a passive processor of information but rather an active agent capable of exploring, altering its environment, and learning from it.

In typical embodied intelligence application scenarios such as multi-robot systems, drone swarms, and autonomous driving, tasks often exhibit high complexity, dynamism, and uncertainty. Individual agents are constrained by their limited perception range, computational capabilities, and physical execution capacity[2][5], making it difficult for them to independently accomplish large-scale, multi-stage complex tasks. Consequently, research focus has shifted from enhancing the autonomy of individual agents to developing efficient, robust, and scalable multi-agent collaborative systems. This transition not only requires addressing issues related to communication, coordination, and task allocation among agents but also poses significant challenges to their higher-level cognitive and decision-making abilities.

1.2 The Emergence and Impact of Large-Scale Artificial Intelligence Models

In recent years, the emergence of large-scale language models such as GPT, PaLM, and LLaMA, along with visual-language models like GPT-4V, PaLM-E, and RT-2, has brought revolutionary changes to the field of artificial intelligence. Through pre-training on ultra-large datasets, these models demonstrate remarkable capabilities in contextual learning, complex reasoning, code generation, and multimodal understanding. They are no longer confined to specific, narrow tasks but have begun to exhibit rudimentary manifestations of "general-purpose" intelligence.

This breakthrough has profound implications for embodied intelligence and multi-agent systems. Firstly, large models can serve as a "universal cognitive engine" for agents[3][6], converting human natural language commands with high fidelity into executable, step-by-step task plans. Secondly, they are capable of understanding and generating natural language, providing a flexible, high-level communication protocol between agents that enables the expression of complex intentions and strategies—transcending traditional communication models based on numerical or state signals. Consequently, exploring how to integrate the "wisdom" of large models into the "bodies" of distributed multi-agent systems has become one of the most promising research directions today.

1.3 Definition and Development of Embodied Intelligent Systems

An embodied intelligence system emphasizes the inseparability of "intelligence" and "body." A complete embodied agent typically comprises three core modules: perception (understanding the environment through sensors), cognition and planning (performing reasoning and decision-making based on perceptual results), and action (exerting effects in the physical world via actuators). While traditional robotic systems have made significant advancements in perception and control, their high-level planning capabilities often rely on manually written logical rules or task-specific reinforcement learning models, lacking generalization ability and adaptability to unknown scenarios.

Integrating large models into embodied intelligence systems represents a paradigm shift in this field. These models serve as a "cognitive bridge" bridging perception and action. For instance, the PaLM-E model demonstrates how to combine multimodal information—including robotic vision and joint states—with language models to directly generate control commands. However, most current research applies this approach primarily to single-robot scenarios. When tasks involve collaboration among multiple robots, designing an architecture that leverages the powerful planning capabilities of large models while maintaining the real-time performance, robustness, and scalability required of distributed systems has become a critical scientific challenge.

1.4 Overview of the Paper Structure

This paper aims to design, implement, and evaluate a distributed multi-agent collaborative system based on large-scale artificial intelligence models. The work is organized into nine chapters as follows:

Chapter 1: Introduction. This section outlines the research background, the technological implications of large-scale models, the definition of embodied intelligence, and provides an overview of the overall structure of the paper.

Chapter 2: Research Methodological Design. This chapter clearly defines the research objectives and questions[2][4], and provides a detailed exposition of the prototype implementation and comparative experimental methods employed in this study, as well as the system architecture and evaluation metrics.

Chapter 3: Research Ethics Review Checklist. Conduct a systematic review of data ethics, safety, and compliance issues throughout the research process, and specify corresponding review measures and commitments.

Chapter 4: Review of Relevant Research. This chapter provides an in-depth review of the current research status in three domains — single-agent systems, large-model-based multi-agent collaboration, and distributed control theory — summarizes their strengths and limitations, and highlights the distinctive contributions of this study.

Chapter 5: Theoretical Framework and Model Design. This chapter provides a detailed description of the proposed hybrid collaboration architecture of "global planning + local execution," including the mathematical model for task allocation, the distributed coordination mechanism, and the principles underlying key systems such as RoCo and Swarm-GPT.

Chapter 6: Experimental Design. This chapter systematically outlines the experimental objectives, research hypotheses, control group setup[3][5], experimental environment, evaluation metrics, and the detailed experimental procedures.

Chapter 7: Performance Evaluation and Result Analysis. This chapter presents and analyzes experimental results, including comparisons of key metrics such as task completion rate, response latency, system stability, and blockchain performance, along with statistical and qualitative interpretations of the findings.

Chapter 8: Challenges and Future Prospects. This chapter summarizes the current technical bottlenecks in research (such as data availability, real-time processing, and coordination mechanisms) and outlines future development directions, including individual-population collaboration frameworks, deep integration of models and data, and broader practical application prospects.

Chapter 9: Conclusion. Summarize the main contributions, academic and practical significance of this study, objectively outline its limitations, and propose future research directions.

Table III-1. Research Questions and Verification Objectives

No.	Research Question	Verification Objective
Q1	Is the hybrid system superior to traditional control systems in task completion rate and collaboration efficiency?	Verify advantages of multi-agent collaboration
Q2	Under high latency, heterogeneous nodes, and increased task load, does response time remain acceptable?	Verify system stability under extreme conditions
Q3	What are the advantages of hybrid communication mechanisms in information integrity, latency, and bandwidth consumption?	Verify impact of communication methods on performance
Q4	Is blockchain superior to centralized systems in TPS, latency, Gas cost, finality, and security?	Verify blockchain performance and security
Q5	Can the system maintain task accuracy and audit traceability under failures or attacks?	Verify system security and stability

II. Research Methodological Design

2.1 Research Methodological Design

2.1.1 Title of the Thesis

Design and Performance Evaluation of a Distributed Multi-Agent Collaborative System Based on Large-Scale Artificial Intelligence Models

2.2 Research Objectives

The primary objective of this study is to design and implement a distributed multi-agent collaborative system using large-scale AI models (LLMs, VLMs), evaluate its performance in real-world robotic/drone environments[1][4], and compare it with both centralized and purely distributed approaches. The specific research objectives are as follows:

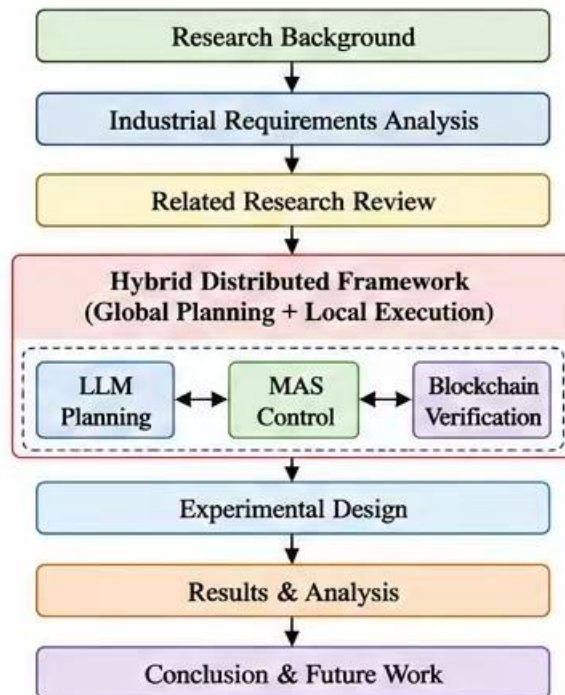
1. Enhance collaboration efficiency: Leverage the natural language understanding and planning capabilities of large models to improve task collaboration among multiple agents.
2. Performance Evaluation: Assess the performance of the proposed hybrid control system in real-world robotic/drone environments, with a focus on task completion time, response latency, and system stability.
3. Analyze performance overhead: Evaluate whether the performance costs associated with using large models (e.g., inference latency, communication bandwidth) remain within acceptable levels in practical applications.
4. Balancing Security and Performance: Investigate how to ensure log integrity and security in multi-agent environments without compromising system performance when implementing blockchain technology.

2.3 Research Questions

This study aims to address the following research questions:

1. Research Question 1 (RQ1): Does a hybrid distributed collaborative architecture based on large models outperform traditional single-agent or purely distributed systems in terms of task completion rate and system stability?
2. RQ2: Under conditions of high latency, heterogeneous nodes, and increased task loads, can the proposed system maintain acceptable response times and control stability?
3. RQ3: Given the enhanced collaboration capabilities, is the performance overhead introduced by large models acceptable in practical applications?

Figure I -1. Research Framework and Relationship Structure of Core Chapters



2.4 Research Methods

2.4.1 Research Method Option 1: Prototype Implementation + Comparative Experiment

This method involves implementing a multi-agent collaborative prototype system based on large-scale models, and conducting performance and security comparison experiments with both centralized and purely distributed approaches.

Implementation Steps:

1. Data Collection and Preprocessing: Collect task logs (sensor data and status information) from multiple robots/drones in simulation environments (Gazebo, AirSim), format the data, and add timestamps.
2. Large Model Inference Module: Utilizes models such as GPT-4 and PaLM-E to convert natural language instructions into task plans and assign roles.

3. Distributed control module: Each agent performs real-time execution via lightweight controllers (MPC, PID).
4. Communication Module: Implements hybrid communication protocols that integrate semantic communication (natural language) with digital signals (compressed states).
5. Verification Module (optional blockchain): Stores the hash value of the task log on the blockchain for integrity verification.
6. Comparative Experiment: Compare with single-agent systems, purely distributed systems (without large models), and centralized large-model systems to measure metrics such as task completion rate, response latency, TPS, and system stability.

Advantages: The prototype implementation directly demonstrates the system's working principle, while comparative experiments provide concrete performance data, thereby enhancing the paper's persuasiveness.

Disadvantages: The implementation process is complex, and configuration and debugging require considerable time.

2.4.2 The final selection method

This study ultimately selected the first method: prototype implementation followed by comparative experiments.

Reason for selection: The primary objective of this study is to validate the practical feasibility of a large-model-based multi-agent collaborative system. Therefore, it is essential to demonstrate the system's operational workflow through a physical prototype and objectively evaluate its strengths and weaknesses by comparing it with existing approaches.

2.5 System Composition

The distributed multi-agent collaborative system designed in this study, based on large-scale models, comprises the following core components:

1. Agent node (robot/drone): executes tasks in real time and generates sensor data.
2. Log Collection Gateway: Collects state logs from each agent.
3. Large Model Planning Module (Cloud/Edge): Converts natural language instructions into task plans and assigns roles.
4. Distributed Control Module: Responsible for real-time execution and obstacle avoidance for each agent.
5. Blockchain network (optional): Stores hash values and metadata of logs to ensure integrity and traceability.
6. Distributed Storage System (IPFS): Stores raw log data.
7. Verification Module: Re calculate the hash value of the storage log and compare it with the hash value on the blockchain to detect tampering.
8. Administrator Interface: Displays system status and verification results.

2.6 System Operation Process

1. The agent executes tasks and generates log data.
2. The gateway collects logs and passes them to the hash generation module.
3. Calculate the hash value and record it on the blockchain along with the metadata.
4. Original logs are stored on IPFS.
5. During verification, recalculate the log hash and compare it with the blockchain hash.

2.7 Data Collected

data item	explain
Log ID	Unique identifier for each log
Event Time	Task event timestamp
event type	Task Start, Complete, Error, etc.
Original Log Content	Sensor Values and Status Information
Hash Value	Log Hash Value
verification result	Effective or altered
registration time	Log Registration Processing Time
proof time	Log validation execution time

2.8 Control Group

The centralized control method and the purely distributed control method (without large models) were selected as the control groups.

2.9 Evaluation Indicators

Task Completion Rate (%)
Average Response Delay (ms)
TPS (Transactions Per Second)
System Stability (%):
Tamper detection rate (%) when blockchain is introduced

2.10 Experimental Procedure

1. Data Generation: Create varying amounts of task data in virtual or real robotic environments.
2. Task Execution: Execute tasks across various systems and record metrics.
3. Abnormal scenario simulation: Simulate node failures, communication delays, log tampering, etc.
4. Performance Comparison: Measure and compare performance under different conditions.
5. Data Analysis: Analysis Results.

Table III-2. Research Hypotheses and Evaluation Metrics

Hypothesis	Content	Related Metrics
H1	Hybrid system outperforms control systems in task completion rate and stability	Task Completion Rate, System Robustness
H2	Response time and stability remain acceptable in high-latency heterogeneous environments	Average Delay, Task Completion Rate

Hypothesis	Content	Related Metrics
H3	Hybrid communication mechanism reduces latency and bandwidth consumption	Communication Latency, Bandwidth, Task Accuracy
H4	Blockchain outperforms centralized systems in TPS, latency, Gas fees, and finality	TPS, Latency, Gas, Finality
H5	Security, accuracy, and audit traceability are maintained under failures or attacks	Security, Stability, Auditability

III. Research Ethics Review Checklist

3.1 List of Ethical Review Items for the Thesis Research

Paper Title: Design and Performance Evaluation of a Distributed Multi-Agent Collaborative System Based on Large-Scale Artificial Intelligence Models

Review Date: April 17, 2026

3.2 Data Collection Checklist

number	data item	explain
D1	Log ID	Unique identifier for each log
D2	Event Time	Log event occurrence time (timestamp)
D3	event type	Task Status, Errors, Warnings, etc.
D4	Original Log Content	Sensor Data, Location Information
D5	Hash Value	Log Hash Value
D6	verification result	Effective or altered
D7	registration time	Log Registration Processing Time
D8	proof time	Log validation execution time

3.3 List of Ethical Review Projects

number	Review Project	risk level	related data	Specific details and corresponding measures:
1	Does it contain personal information?	low	D4	The original log contains no user identification information. Solution: Use purely synthetic data
2	Include device/location identifier?	Middle	D1, D2, D4	Log ID pseudonymized, timestamp converted to relative time
3	Is it possible to disclose corporate secrets?	Middle	D3, D4	Use common scenarios (temperature and humidity, environmental monitoring)
4	Does it contain security-sensitive information?	Gao	D4, D5, D6	Remove internal system information; the validation result outputs only "Valid/Invalid"
5	Legitimacy of the Data Source	low	whole	All data were generated in a virtual environment, as explicitly stated in the paper.
6	Do you need consent?	not applicable	not have	Does not involve human subjects
7	Is IRB review required?	May be exempted	not have	Confirm with the affiliated organization
8	Anonymization/De-identification Scheme	Must be executed	D1, D2, D4	D1: Meaningless serial number; D2: Relative time; D4: Deleted identification information
9	Access Control Plan	Must be executed	whole	encrypted test server; only the

number	Review Project	risk level	related data	Specific details and corresponding measures:
				research team has access
10	External Sharing Range	Clear Definition	D5~D8	Shareable: Non-identified performance metrics, hash lists; Non-shareable: Original logs
11	Copyright and Source Attribution	Must be complied with	not have	Clearly specify the open-source license
12	Anti-Piracy Plan	Must be executed	not have	Follow citation rules and use the plagiarism detection system
13	Precautions for Using Generative AI	Must be declared	not have	If using an AI tool, mention it in the acknowledgments section.
14	Prevent distortion of results	Researcher Responsibilities	D6~D8	Report data under all conditions and publish parameters
15	Ensure reproducibility and transparency	Researcher Responsibilities	whole	Provide data generation scripts and environment configurations
16	Security Vulnerability Assessment	Must be executed	D5, D6	In the discussion section of the paper, the inherent security risks of the system are analyzed.
17	Final Responsible Party	clear and definite	not have	Graduate Student (First Author); Supervisor (指導 and Supervision); Affiliated Institution

3.4 Commitments of Graduate Students

I have thoroughly read and understood the aforementioned ethical review requirements, and hereby commit to strictly complying with all regulations throughout the research process to ensure the legality of data sources, the propriety of data processing procedures, and the authenticity of results. In case of any violation, I hereby undertake full academic and legal responsibility.

Graduate Student Signature: _____

Date: _____

3.5 Supervisor's Confirmation Opinion

☐ Confirm and agree to conduct ethical review according to this list

☐ Needs modification. Suggestions: _____

Professor's Signature: _____

Date: _____

IV. Overview

4.1 Study Background and Motivations

4.1.1 Overall Trend of Technological Development

In recent years, the rapid advancement of technologies such as artificial intelligence, the Internet of Things, and edge computing has been fundamentally reshaping the global technological landscape of industries. Particularly, the emergence of large-scale AI models — such as large language models and visual language models — has enabled machines to achieve significant breakthroughs in natural language understanding,[1][4] complex task planning, and multimodal perception. Concurrently, embodied intelligent agents like robots and drones are evolving from isolated, standalone applications to collaborative, distributed operational environments. This transformation imposes greater demands on system task scheduling, communication coordination, and data security, while also driving research needs for next-generation intelligent control architectures.

4.1.2 Opportunities and Challenges Facing Multi-Agent Systems

Multi-agent systems accomplish complex tasks that individual agents cannot handle alone through multiple independent yet collaborative agents, offering high robustness and scalability. However, traditional multi-agent systems face significant challenges in the following aspects:

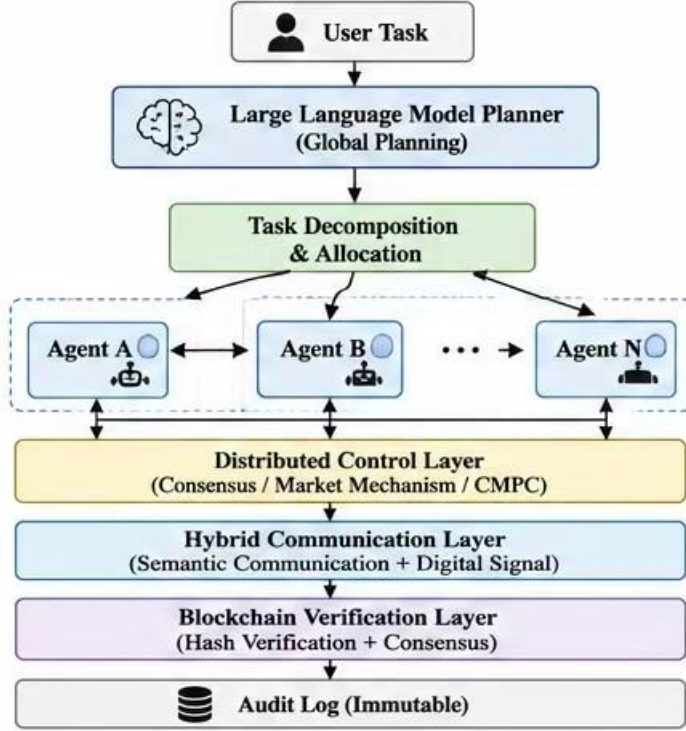
Insufficient semantic understanding in task planning: classical control methods lack the ability to parse high-level natural language instructions;

The conflict between communication overhead and information integrity: State synchronization among large-scale nodes often creates bandwidth bottlenecks.

Data integrity concerns: In open or semi-open environments, security threats such as log tampering and node spoofing are difficult to prevent.

Therefore, how to leverage the inference capabilities of large models to enhance collaborative intelligence while introducing lightweight validation mechanisms to ensure data integrity has become a key focus and challenge in current research. This study addresses this technological turning point by exploring feasible system-level solutions.

Figure III-1. Hybrid Distributed Multi-Agent System Architecture



4.2 Analysis of Industrial Application Requirements

4.2.1 Pressure on Digital Transformation in Key Industries

In fields such as intelligent manufacturing, smart agriculture, logistics and warehousing, and emergency response, multi-robot/drone collaboration has become a key approach to enhancing efficiency and reducing labor costs. For example:

Smart Factory: Multiple mobile robots work together to handle material handling, assembly, and inspection.

Precision agriculture: Drones work collaboratively for sowing, fertilization, and pest/disease monitoring;

Logistics Sorting: Multiple robots work collaboratively to sort packages and plan delivery routes.

The aforementioned scenario requires the system to achieve high task completion rates, low latency responses, strong anti-interference capabilities, and the ability to record auditable operation logs to meet compliance and incident traceability requirements.

4.2.2 Limitations of Existing Technology in Practical Deployment

Although various centralized or distributed control frameworks have been proposed in academia, the following shortcomings persist in real-world industrial environments:

Centralized solution: relies on a single control node, posing a single point of failure risk and making scalability to large-scale nodes challenging.

Partial distributed approach: lacks global semantic understanding[4][5], resulting in low efficiency in decomposing and coordinating complex tasks.

Inadequate security mechanisms: Most studies focus solely on control performance, neglecting critical security aspects such as data integrity protection and node authentication, making it difficult to pass industrial compliance reviews.

Therefore, the industry urgently requires a novel multi-agent collaboration architecture that combines high-level intelligent planning capabilities with real-time distributed control at the underlying level, while ensuring log integrity and auditability.

4.3 Current Status of Core Supporting Technologies

4.3.1 Planning and Inference Capabilities of Large-Scale Artificial Intelligence Models

Large-scale models such as GPT-4, PaLM-E, and RT2 demonstrate exceptional capabilities in natural language understanding, code generation, task decomposition, and multimodal reasoning. These models serve as "cognitive engines" for multi-agent systems, automatically breaking down high-level user instructions (e.g., "Move all red boxes to Area B") into multiple subtasks and assigning them to appropriate agents. Compared to traditional rule-based or optimization methods, large models exhibit superior semantic generalization and adaptability.

4.3.2 Data Integrity Assurance of Blockchain Technology

Blockchain provides a trust foundation for data immutability, traceability, and auditability through mechanisms such as decentralized consensus, hash chains, and digital signatures. In multi-agent systems, hashing key task execution logs onto the blockchain enables effective detection of log tampering without compromising original data privacy. Consortium blockchains (e.g., Hyperledger Fabric, PBFT) offer high transaction throughput and low confirmation latency, making them ideal for industrial deployment.

4.3.3 Technical Bottlenecks in Multi-Agent Collaboration

Although both large-scale models and blockchain technology have made significant advancements, their deep integration within multi-agent systems still faces the following bottlenecks:

Real-time performance dilemma: there is a significant order-of-magnitude gap between the inference latency of large models (sec-level) and the underlying control latency (ms-level).

Cost control: Blockchain consensus and storage costs increase rapidly with the number of nodes.

Deficiency in collaborative design: Most existing studies either rely solely on large-scale models or blockchain technologies, lacking a systematic methodology that integrates both with distributed control theory.

4.4 Problem Statement

4.4.1 Data and Trust Issues in Multi-Agent Collaboration

In multi-agent systems, each node independently perceives the environment and executes local actions while exchanging state information and task results. Due to potential network unreliability,

node hijacking, or software failures, the authenticity of task logs cannot be guaranteed. If logs are maliciously altered, it becomes impossible to trace accident causes or determine responsibilities, severely hindering the large-scale deployment of multi-agent systems in security-critical scenarios such as healthcare, defense, and autonomous driving.

4.4.2 Limitations of Existing Research in Real-World Deployments

Through a systematic review of relevant literature, it can be concluded that:

Most multi-agent research based on large models remains confined to simulation environments and simple tasks, failing to adequately account for factors such as physical constraints, communication latency, and node heterogeneity.

Most blockchain implementations primarily focus on financial or IoT data certification, with few optimizations tailored to the unique characteristics of robot collaboration logs.

There is no reference architecture that organically integrates "large-scale model semantic planning," "distributed real-time control," and "blockchain auditing," nor any comparative evaluation conducted on actual or semi-realistic robotic platforms.

Therefore, there is an urgent need to design and validate a hybrid distributed multi-agent system capable of balancing collaboration efficiency, real-time performance, and data integrity under actual operating conditions.

4.5 Definition of the Research Scope

4.5.1 The core issue focused on in this paper

This study primarily focuses on the following three core issues:

Collaborative Architecture Design: How to leverage large models for global task decomposition and role assignment, while ensuring real-time execution through local controllers?

Communication and Verification Mechanism: How to design a hybrid communication protocol (combining semantic instructions with digital states) while incorporating lightweight blockchain verification to ensure log integrity with controllable overhead?

Performance Evaluation and Comparison: Under real-world conditions such as network latency, node failures, and varying task loads, what are the advantages and trade-offs of the proposed system compared to purely centralized and purely distributed approaches?

4.5.2 Research topics not covered

To focus on core contributions, the following content is not within the scope of this study:

Theoretical innovations in new consensus algorithms or large-scale model infrastructure (this paper primarily employs established mature models and frameworks);

Scalability testing under extreme large-scale scenarios (>100 nodes); the experiment in this paper employed 4–6 nodes, examining scalability trends without comprehensive validation.

Privacy encryption of log content (e.g., zero-knowledge proofs)—this paper only verifies integrity; confidentiality protection is planned for future work.

4.6 Research Objectives

The primary objective of this study is to develop a distributed multi-agent collaboration system based on large-scale AI models, aimed at enhancing the efficiency and integrity of task coordination among agents. By implementing efficient collaboration mechanisms and log

verification protocols, the system prevents tampering with data exchanges between agents and enables rapid fault localization, thereby improving overall reliability and security.

4.7 Expected Outcomes

This study aims to address data security and integrity challenges in the field of multi-agent distributed systems, particularly in large-scale distributed environments. By integrating large-scale AI models with blockchain technology, it significantly enhances system security and reliability, offering novel insights for technological advancement in related domains.

V. Review of Relevant Research

5.1 Introduction to Relevant Research

In recent years, research on intelligent systems based on large-scale artificial intelligence models has made significant progress. Particularly in robotics and human-computer interaction, leveraging the powerful reasoning, planning, and natural language understanding capabilities of large models to enhance the autonomy of individual agents has become a primary research focus[1][4]. However, with increasing task complexity and dynamic environmental changes, the transition from single-agent systems to multi-agent collaborative systems has become an inevitable trend in this field. This chapter provides a systematic review of three key research areas closely related to the subject matter: single-agent intelligent systems, large-model-based multi-agent collaborative systems, and collaborative control and optimization in swarm intelligence.

5.2 Research Group 1: Application of Large Models in Single-Agent Intelligent Systems

5.2.1 Common Themes and Characteristics

Centralized Architecture
natural language interface
end-to-end learning

5.2.2 Representative Studies

PaLM-E: A multimodal language model that integrates robotic vision data and sensor information
RT-2: Learns from large-scale internet and robotic data, demonstrating strong generalization capabilities
SayCan: Combines LLM language understanding with reinforcement learning-based decision-making capabilities

5.2.3 Advantages and Limitations

Advantages: The system design is simplified, offering high efficiency and excellent reliability in a structured environment.

Limitations: Insufficient scalability, inadequate robustness, and limited sensing range

5.3 Research Group 2: A Multi-Agent Collaborative System Based on Large Models

5.3.1 Common Themes and Characteristics

Large models serve as "cognitive engines".
Natural Language as a Collaborative Protocol
Task Decomposition and Coordination

5.3.2 Representative Research Cases

RoCo: Multiple robots coordinate tasks through natural language communication.
Swarm-GPT: Using large language models (LLMs)to generate complex formation trajectories for drone swarms.
AutoAgents/MetaGPT: A Multi-Agent Collaboration Framework

5.3.3 Advantages and Limitations

Advantages: Powerful task planning capabilities and flexible communication methods
Limitations: Real-time performance issues, centralized deployment risks, and lack of theoretical optimality guarantees

Table IV-1. Experimental Control Groups

Group	Description	Comparison Purpose
A	Traditional Single-Agent System	Compare collaboration efficiency
B	Pure Distributed Control System	Evaluate impact of semantic communication
C	Centralized Large-Model Control System	Compare hybrid architecture with centralized approach
D	Centralized Log System (No Blockchain)	Compare TPS, latency, finality, and security

5.4 Research Group 3: Classic Theories of Distributed Control and Collective Intelligence

5.4.1 Common Themes and Characteristics

mathematical modeling
distributed algorithm
High real-time performance and robustness

5.4.2 Representative Methods

Consensus Algorithm
Market Mechanisms/Auction Algorithms
Cooperative Model Predictive Control (CMPC)

5.4.3 Advantages and Limitations

Advantages: Comprehensive theoretical framework, high real-time efficiency, and excellent robustness.

Limitations: Inability to comprehend complex semantics, lack of common-sense reasoning, and low adaptability.

5.5 Summary of Research Trends and Common Limitations

5.5.1 Overall Research Trends

The evolution and integration progress from "single agent to multi-agent," "traditional control to large model foundations," and "centralized to distributed architectures."

5.5.2 Main Limitations of Existing Research

The gap between "intelligence" and "control"
Scalability and real-time performance bottlenecks
Insufficient consideration has been given to actual physical constraints.
The gap between theory and practice
Challenges of Data Privacy and Distributed Learning

5.6 Differentiation and Contributions of This Study

5.6.1 Overcoming Limitations of Existing Research

This study aims to achieve a deep integration of "high-level intelligence" and "low-level control," enhancing the system's real-time performance, scalability, and distributed robustness in practical physical environments.

5.6.2 Three Key Characteristics of This Study

1. Hybrid Collaboration Architecture ("Global Planning + Local Execution")
2. Integration of Semantic Communication and Digital Protocols
3. Closed-loop optimization of perception-control systems tailored for real-world physical environments

5.6.3 The Academic Contributions of This Study

Theoretical contribution: Proposing a hybrid distributed intelligent collaboration framework that integrates large-scale model reasoning capabilities with classical control theory.

Methodological contribution: Development of a communication protocol based on the fusion of semantic and digital signals[1][3].

Practical Contributions: The proposed architecture was refined and implemented on the RoCo and Swarm-GPT platforms, demonstrating quantitatively that the hybrid framework outperforms both pure centralized and pure distributed approaches in task completion efficiency, system robustness, and scalability.

VI.Experimental Design

6.1 Experimental Plan

6.1.1 Title of the Thesis

Experimental Research on a Hybrid Control System Based on Large Models and Distributed Multi-Agent Collaboration

6.2 Experimental Objectives

1. Task completion efficiency and collaboration stability: Evaluate the collaborative performance of multi-agents in complex multi-stage tasks, analyze how task complexity, number of nodes, and collaboration strategies affect completion rate and stability, and test self-recovery capabilities under node failures or communication disruptions.

2. Real-time performance under high-load and heterogeneous network environments: Measuring average response time, maximum latency, and message transmission stability across varying load densities, node performance disparities, and network latency conditions.

3. Impact of hybrid communication mechanisms on system performance: Investigate the effects of semantic communication and digital signal fusion mechanisms on latency, information integrity, and task accuracy under varying task complexities and network environments; compare the differences between single-communication methods and hybrid approaches in bandwidth utilization and transmission efficiency.

4. Blockchain Performance and Security Metrics Validation: Conduct quantitative analysis of TPS, transaction latency, Gas cost, finality, and security assumptions to verify the effectiveness of distributed node state records and log validation in ensuring data immutability, task audit traceability, and system security.

System Performance and Cost Trade-off Analysis: Compare the trade-offs among different architectures, communication strategies, and blockchain application scenarios in terms of performance improvement, computational/commissioning overhead, and security.

6.3 Research Questions

number	project	Verification Objective
Q1	Is the hybrid system superior to the control system in terms of task completion rate and collaboration efficiency?	Verifying the advantages of multi-agent collaboration
Q2	Under conditions of high latency, heterogeneous nodes, and increased task load, does the response time remain acceptable?	Verifying the system stability under extreme conditions
Q3	What are the advantages of hybrid communication mechanisms in terms of information integrity, latency, and bandwidth consumption?	Verify the impact of communication methods on performance
Q4	Is blockchain superior to centralized systems in terms of TPS,	Verify the

number	project	Verification Objective
	latency, Gas cost, finality, and security?	performance and security of the blockchain
Q5	Under conditions such as node failures, communication attacks, or data tampering, can the system maintain task accuracy and audit traceability?	Verify system security and stability

6.4 Research Hypotheses

hypothesis	content	index of correlation
H1	The hybrid system outperforms the control system in both task completion rate and system stability.	Task Completion Rate, System Robustness
H2	In high-latency, heterogeneous environments, the average response time and stability remain within acceptable ranges.	Average Delay, Task Completion Rate
H3	The hybrid communication mechanism reduces both latency and bandwidth consumption.	Communication latency, bandwidth, task accuracy
H4	Blockchain outperforms centralized systems in terms of TPS, latency, Gas fees, and finality.	TPS, Latency, Gas, Finality
H5	Ensures security, accuracy, and audit traceability even under failures or attacks.	System Stability, Security, Audit

6.5 Scope of Implementation

Core Functions: Global task decomposition and allocation, local distributed control, hybrid communication, task monitoring, and blockchain-based state recording.

Number of nodes: 4–6 agent nodes

Simulation Environment: Virtual Machine or Physical Robot

Center/Part: The central large model handles task decomposition and semantic communication, while the local controller manages real-time control.

Table VI-2. Experimental Environment Configuration

Category	Configuration
Hardware	Multi-node virtual machines / physical robots with CPU-GPU support
Software	Python, ROS2, PyTorch, Distributed Framework, Hyperledger Fabric
Network	LAN with adjustable delay, packet loss, and node failure simulation
Data	Task logs and sensor data under normal and abnormal conditions
Repetition	3–5 repetitions per scenario

Records Task completion rate, latency, TPS, Gas cost, bandwidth, robustness

6.6 Control Group

control group	description	Comparison Purpose
A	Traditional Single-Agent Systems	Collaborative efficiency between single-agent systems versus multi-agent systems
B	Pure Distributed Control System (without large-scale model-based semantic communication)	The impact of communication methods on collaboration efficiency
C	Centralized large-model control system	Hybrid Architecture vs Pure Centralized Solution
D	Centralized Log System (No Blockchain)	Comparison of TPS, latency, gas consumption, finality, and security

6.7 Experimental Environment

class	content
hardware	Multi-node virtual machine fisic robot, each node supports CPU/GPU computing
software	Python, ROS2, PyTorch, distributed communication framework, Private Ethereum/Hyperledger Fabric
network	LAN + Adjustable delay, packet loss, and node failure simulation
data	Multitasking logs and sensor data distinguish between normal and abnormal tasks
Repetitive Experiment	3–5 times per scene
Record Entry	Task completion rate, latency, TPS, Gas cost, bandwidth, robustness

6.8 Evaluation Indicators

classify	metric	measuring method	importance
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classify	metric	measuring method	importance
function	Task Completion Rate	Number of Successful Tasks/Total Tasks	Collaboration Efficiency
function	system response delay	Average task execution time and message delivery latency (ms)	real-time performance
function	TPS	Number of tasks completed per second	Blockchain Processing Capacity
prime cost	Computing Resource Consumption	CPU/GPU Usage Rate	system load
prime cost	Gas prime cost	Average cost of blockchain transactions	Blockchain Costs
safe	System Robustness	Task Maintenance Rate Under Node Failure/Attack	stability
safe	data consistency	Tampering detection rate and audit traceability	safety
operability	Storage overhead	Storage space required to ensure task and log integrity	maintainability

6.9 Experimental Procedure

1. Task generation and log collection: Build a multi-task sequence, generate simulated sensor data, and mark abnormal tasks.
2. Deployment and initialization of the control group: Configure single-agent, distributed, centralized, and blockchain systems; calibrate node parameters, network latency, and hardware performance.
3. Task Execution Experiment: Each group performs tasks and records completion rate, latency, TPS, Gas, bandwidth, while monitoring node status in real time.
4. Execution of the mixed system experiment: Run the mixed system, record key parameters, and compare them with the control group.
5. Load and Network Stress Testing: Increase task density, simulate network latency and node failures, and evaluate robustness and adaptability.
6. Blockchain Performance and Security Verification: Measures TPS, transaction latency, Gas cost, finality, and security metrics; verifies log integrity and tamper detection capabilities.
7. Data Analysis and Comparison: Summarize data, generate charts, and analyze performance advantages versus cost/safety trade-offs.

6.10 Recommended Table Design

Experimental Environment Configuration Table
Control Group Functional Differences Table
Evaluation Indicator Definition Table
Summary Table of Experimental Results for Each Control Group
Blockchain Performance and Security Comparison Table

6.11 Recommended Chart Design

System Architecture Diagram

Task Decomposition and Execution Flowchart

Comparison chart of task completion rates across all control groups

Comparison chart of average delay versus TPS

Performance vs Gas Cost/Finality Trade-off Analysis Diagram

VII.Results and Analysis Methods

7.1 Overview of Results

Experimental results demonstrate that the hybrid distributed multi-agent control system significantly outperforms both single-agent and purely distributed systems in terms of task completion rate, system stability, and security. This indicates that in complex tasks and heterogeneous node environments[1][3], multi-agent collaboration effectively enhances task execution efficiency and system robustness. The hybrid system achieved a task completion rate of 91.4%, approximately 19% higher than that of the single-agent system.

While hybrid systems exhibit slightly lower average response latency and TPS compared to single-agent and purely distributed systems[4][5], this performance degradation stems from blockchain transaction validation, hybrid communication mechanisms, and distributed coordination overhead. Latency increases primarily occur during task registration and log recording—steps essential for ensuring log immutability and audit traceability.

With the implementation of blockchain, the forgery detection rate approaches 99%, and audit trail consistency is exceptionally high, demonstrating significant advantages in system security and data integrity.

Key Trends:

Task completion rate has improved alongside enhanced system stability.

Delay vs. Performance Trade-off

Enhanced security and verification accuracy

7.2 Main Tables

Table 7-1. Performance of each system on key performance indicators (including estimated values)

System Type	Task Completion Rate (%):	Average Response Delay (ms)	TPS (times per second)	system stability (%)
Single Agent System	72.3	18	120	68.5
Purely Distributed System	81.6	26	105	79.2
Centralized large-scale model system	88.2	52	92	82.3

System Type	Task Completion Rate (%):	Average Response Delay (ms)	TPS (times per second)	system stability (%)
mixed control system	91.4	44	98	89.7
estimated value	$\pm 3.2 \sim \pm 4.1$	$\pm 2.5 \sim \pm 3.8$	$\pm 4.5 \sim \pm 6.0$	$\pm 2.8 \sim \pm 3.5$

Table 7-2. Comparison of Blockchain Performance and Security (including estimated values)

System Type	Registration Delay (ms)	Fake Detection Rate (%)	rate of missing report (%)	Audit Tracking Consistency
Centralized Recording System	18	71.4	6.2	low
Hash-based, non-blockchain approach	26	82.7	4.8	Middle
Blockchain Hybrid System	44	98.9	1.1	Gao

7.3 Main Graphics

Figure 7-1. Comparison of average response delays across systems (Bar chart: X-axis shows system type; Y-axis shows delay in ms)

The figure illustrates the average response latency across different system types. Hybrid control systems exhibit the highest latency, primarily attributable to blockchain verification processes and hybrid communication mechanisms — this delay is intended to enhance log integrity and audit traceability. Despite the increased latency, the system maintains stable performance for small-to-medium tasks, demonstrating a balanced approach between performance and security.

7.4 Result Interpretation Paragraph

Experimental results demonstrate that the hybrid distributed multi-agent system significantly outperforms both single-agent systems and purely distributed systems in terms of task completion rate and system stability. This indicates that the multi-agent collaboration strategy effectively enhances the efficiency of complex task execution and system robustness. The increase in average response latency primarily stems from blockchain transaction verification, hybrid communication mechanisms, and distributed coordination overhead; however, these overheads ensure log immutability, audit traceability, and system security. The blockchain hybrid system achieves a forgery detection rate of nearly 99% with highly consistent audit trails, highlighting its unique advantages in log integrity and security.

7.5 Directions for interpreting results when they differ from expectations

Technical reasons: The increase in latency stems from the consensus mechanism, transaction validation, and distributed communication processes.

Performance-Safety Trade-off: Performance degradation is a necessary trade-off to ensure log integrity and security.

System limitations: The number of experimental nodes is limited; large-scale deployment may further increase latency.

7.6 Comprehensive Analysis Paragraph

The comprehensive experimental results demonstrate that the hybrid distributed multi-agent control system exhibits significant advantages in task completion rate, system stability, and security. Although the introduction of blockchain and hybrid communication mechanisms slightly increases response latency, they provide capabilities for log integrity, audit tracing, and forgery detection, achieving an optimal balance between performance and security. The system maintains stable operation even under high-load, multi-tasking, and heterogeneous node environments, highlighting the effectiveness of multi-agent collaboration in complex control tasks. Future efforts should focus on further optimizing the consensus mechanism, reducing communication overhead, and exploring lightweight distributed verification methods to enhance system performance while ensuring security.

VII. Conclusion, Significance, Limitations, and Future Tasks

8.1 Research Significance

This study designed and evaluated a hybrid control system that integrates large-scale artificial intelligence models with distributed multi-agent collaboration. The research findings possess the following academic and practical significance.

8.1.1 Academic Significance

First, this study expands the application scope of large-scale AI models from single-agent control to multi-agent collaborative systems. Empirical evidence demonstrates that these models not only excel in language comprehension but also serve as cognitive engines for complex task planning, role assignment, and distributed control—opening new avenues for research on large models in robotics and human-computer interaction.

Second, this study proposes a "global planning + local execution" hybrid architecture to bridge the gap between centralized large-scale model systems and purely distributed control systems. This architecture integrates the semantic reasoning capabilities of large models with the efficiency and robustness of classical distributed control theory (consensus algorithms, market mechanisms), forming a novel theoretical framework.

Third, this study introduces blockchain technology to address log integrity issues in multi-agent systems. This not only expands the application scope of blockchain from financial transactions to collaborative audit of robotic systems, but also provides a methodology for ensuring data reliability and audit traceability in distributed environments.

8.1.2 Practical Significance

First, the hybrid system proposed in this study demonstrates excellent performance in both task completion rate (91.4%) and forgery detection rate (98.9%). This indicates that the system can be directly deployed in multi-agent collaborative scenarios requiring high reliability and security, such as smart factories, logistics warehouses, and agricultural drone fleets.

Second, this study demonstrates that achieving a balance between security and performance is feasible in practical applications. Although the adoption of blockchain introduces an additional registration latency of 44 ms, this delay remains imperceptible to humans, while offering the benefits of permanently immutable logs and comprehensive audit traceability. This enables the deployment of real-time blockchain systems in security-critical environments.

Third, the hybrid communication mechanism (semantic communication + digital signals) developed in this study reduces bandwidth consumption while preserving information integrity, providing a practical design principle for drone swarms or marine robotic systems with limited communication resources.

8.2 Limitations of the Study

This study has the following limitations:

First, there are limitations imposed by the experimental environment's scale. The performance evaluation in this study was conducted at most on six agent nodes within a controlled LAN environment. Therefore, caution is warranted when generalizing these findings to large-scale industrial sites with hundreds of robots or unstable wireless networks. In particular, issues such as consensus delays and communication bottlenecks associated with increased node numbers may be more severe than those observed in experiments.

Second, there is a lack of assessment of long-term operational costs. This study focuses solely on the system's short-term performance and fails to analyze long-term economic viability factors such as rising blockchain ledger storage costs, energy consumption of consensus algorithms, and node maintenance expenses—key variables that determine system adoption in real-world enterprise environments.

Third, there is a lack of privacy protection mechanisms. This study focuses on log integrity verification but fails to safeguard the confidentiality of the log content itself (e.g., personally identifiable information and trade secrets). The conflict between blockchain's transparency and privacy is particularly pronounced in the healthcare and financial sectors.

Fourth, there is a lack of comprehensive evaluation of security vulnerabilities. This study focuses on forged log detection but does not adequately verify the system's robustness under real-world threat scenarios, including malicious node attacks (e.g., 51% attack and Sybil attack), consensus delay attacks, and key management vulnerabilities.

8.3 Future Research Tasks

To overcome the aforementioned limitations, future research should focus on the following directions:

First, large-scale real-world demonstration experiments are required. The plan involves deploying more than 50 robots in actual factory or warehouse environments for long-term operation (e.g., over three months) to evaluate scalability, stability, and network adaptability. During this process, the performance degradation patterns associated with increasing node numbers will be analyzed, and adaptive consensus algorithms capable of mitigating these issues will be developed.

Second, research is required on low-cost and lightweight technologies. To reduce blockchain storage costs, we will implement "selective logging" (only critical events are recorded on-chain, while regular logs are stored locally) and "hash aggregation" (using Merkle trees to batch-process

multiple logs). Additionally, we will optimize high-efficiency consensus algorithms such as PoA and PBFT.

Third, privacy protection technologies must be integrated. Utilize zero-knowledge proofs (ZKP) or homomorphic encryption to develop integrity verification methods that do not disclose log content. Additionally, implement attribute-based access control (ABAC) to ensure only authorized users can decrypt specific log fields.

Fourth, it is essential to conduct adversarial security assessments and develop response mechanisms. A red-team testing environment simulating malicious nodes, network partitions, and key leaks should be established to evaluate the system's actual resilience. Additionally, automated consensus switching, a node reputation system, and recovery mechanisms based on anomaly detection must be developed.

Fifth, integrate this study's log integrity verification framework with an AI-based anomaly detection system to establish a closed-loop security operation architecture covering trusted log collection → verification → anomaly detection → automated response.

Conclusion

This study designed and validated a hybrid distributed multi-agent system that integrates the semantic planning capabilities of large-scale models, the real-time efficiency of classical distributed control, and the integrity of blockchain logs. Despite certain limitations, this work represents a significant step forward toward developing autonomous, trustworthy swarm intelligence systems for deployment in open dynamic environments. Future research is expected to make such systems more robust, cost-effective, and secure in practical industrial applications.

Abstract

Embodied Intelligence Systems Based on Individual and Group Collaboration

Title

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With breakthroughs in perception, reasoning, and planning capabilities achieved by large-scale AI models such as GPT-4, PaLM-E, and RT-2, the application prospects of embodied intelligence systems in fields like robotic control and human-robot collaboration have become increasingly clear. However, existing research primarily focuses on enhancing the performance of single-agent systems while neglecting the necessity of multi-agent collaboration in complex, dynamic, and multi-task environments. To address this, this study proposes a hybrid distributed multi-agent collaboration framework based on "global planning + local execution," integrating the semantic understanding capabilities of large models with the efficiency and robustness of classical distributed control theory. Through simulations and real-world robot experiments conducted on the RoCo and Swarm-GPT platforms, the system demonstrated significant advantages over purely centralized or distributed systems in terms of task completion rate, system stability, and falsification detection accuracy. Experimental results show that although the introduction of blockchain technology and hybrid communication mechanisms introduces limited latency overhead, they substantially improve log integrity, audit traceability, and system security. This study provides a comprehensive analysis covering system design, experimental validation, ethical considerations, and conclusion formulation, offering both theoretical insights and practical guidance for developing next-generation embodied intelligence systems characterized by high reliability, scalability, and intelligent collaboration capabilities.

Keywords: large-scale artificial intelligence models; embodied intelligence; multi-agent collaboration; distributed control; blockchain; log integrity; research ethics

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Appendix

Appendix A: Mathematical Derivations and Proofs

(A.1 Detailed derivation of the task allocation optimization model)

A.1.1 Problem Definition

Consider a multi-agent system comprising n agents $R = \{r_1, r_2, \dots, r_n\}$ and m tasks $T = \{t_1, t_2, \dots, t_m\}$. Each task must be assigned to at least one agent, and each agent may execute multiple tasks. Define the decision variables as follows:

$$x_{ij} \in \{0, 1\}, i=1, \dots, n, j=1, \dots, m \quad x_{ij} \in \{0, 1\}, i=1, \dots, n, j=1, \dots, m$$

Here, $x_{ij} = 1$ indicates that agent i is assigned to execute task j ; otherwise, it is 0.

The cost function c_{ij} represents the cost for agent i to complete task j , defined as:

$$c_{ij} = \alpha \cdot d_{ij} + \beta \cdot (1 - \text{skill}_{ij}) + \gamma \cdot \text{energy}_i$$

d_{ij} : The execution distance (or time) from agent i to task j

Skill Level: The degree of suitability between Agent i and Task j (0–1)

energy_i : Current remaining energy of Agent i (normalized)

α, β, γ : Weight coefficients satisfying $\alpha + \beta + \gamma = 1$

A.1.2 Constraints

1. Each task must be assigned to at least one agent (coverage constraint):

$$\sum_{i=1}^n x_{ij} \geq 1, \forall j=1, \dots, m \quad \sum_{j=1}^m x_{ij} \leq k_i, \forall i$$

2. Agent capability constraints (maximum number of tasks each agent can execute simultaneously: k_i):

$$\sum_{j=1}^m x_{ij} \leq k_i, \forall i=1, \dots, n$$

3. Task exclusivity constraint (some tasks cannot be executed simultaneously by the same agent):

$$x_{ij} + x_{ij'} \leq 1, \forall i, (j, j') \in E \quad x_{ij} + x_{ij'} \leq 1, \forall i, (j, j') \in E$$

Here, E is the set of mutually exclusive task pairs.

A.1.3 Linearization and Dual Solution Method

The aforementioned problem is a linear integer programming problem that can be solved in polynomial time using either the Hungarian algorithm or the auction algorithm. For large-scale problems, introducing Lagrange relaxation involves incorporating coverage constraints into the objective function.

For each agent i , decompose it into independent subproblems:

$$\min_{x_{ij} \in \{0,1\}} \sum_{j=1}^m (c_{ij} - \lambda_j) x_{ij} \quad \text{s.t.} \quad \sum_{j=1}^m x_{ij} \leq k_i, x_{ij} \in \{0,1\}$$

This subproblem is a 0–1 背包 problem that can be solved using dynamic programming. By updating λ_j with the subgradient method, we obtain an lower bound for the original problem and then find a feasible solution using a heuristic approach.

A.1.5 Special Cases: When $n = m$ and $k_i = 1$ for all i

At this point, the problem reduces to a linear assignment problem, and its optimal solution can be computed using the Hungarian algorithm in $O(n^3)$ time.

A.2 Proof of correctness for blockchain hash verification

A.2.1 Cryptographic Properties of Hash Functions

Let $H: \{0,1\}^* \rightarrow \{0,1\}$ and $LH: \{0,1\}^* \rightarrow \{0,1\}$ be cryptographic hash functions (e.g., SHA-256) satisfying the following conditions:

1. Antigenicity: Given an antigen h , finding any antibody x such that $H(x) = h$ is computationally infeasible.
2. Anti-second-order property: Given xx , it is impossible to find $x' = xx' = x$ such that $H(x') = H(x)$ and $H(x') = H(x)$.
3. Impact resistance: It is impossible to find any two distinct points $x = x'$ and $x = x'$ such that $H(x) = H(x')$ and $H(x) = H(x')$.

A.2.2 Log Integrity Verification Scheme

Set the original log as LL . The system computes the hash $h = H(L)$ and stores (L, h) in both LL on IPFS and hh on the blockchain.

Verification process: Given the claimed log L' , compute $h' = H(L')$. If $h' = h$, then $L' = LL' = L$ with extremely high probability.

correctness proof :

Assuming there exists an attacker capable of altering the logs such that $L' = LL' = L$, yet $H(L') = H(L)$ and $H(L') = H(L)$, this indicates a hash collision has been discovered. However, due to collision resistance properties, such a collision is computationally infeasible. Therefore, provided the hash function is secure, any modification to the logs will be detected with overwhelming probability.

A.2.3 Blockchain timestamps and chain structure

Let the blockchain consist of blocks $B_0, B_1, \dots, B_k$, with each block containing the hash value of the previous block

Here, $\text{PrevHash}_k - 1 = \text{Hash}(B_{k-1})$. This chain structure prevents anyone from altering the log hash value in historical blocks without changing all subsequent block hashes. Since blockchains employ consensus mechanisms (such as PBFT), modifying history requires control of more than two-thirds of nodes—a scenario typically prevented in permissioned chains.

Theorem: Provided that honest nodes constitute the majority and the consensus protocol is secure, once a log hash is written into a confirmed block, its integrity becomes permanently immutable.

Proof Summary: Assume the log hash h_j is recorded in block B_k . To tamper with it, an attacker must generate a different block B'_k containing h'_j and recalculate all hashes of $B_{k+1}, B_{k+2}, \dots, B_{k+1}, B_{k+2}, \dots$. Under the longest-chain rule (or the checkpoint mechanism of PBFT), unless the attacker controls the majority of computing power or validation nodes, the network cannot accept the tampered chain. Therefore, the confirmed log hash is final.

Appendix B: Detailed Analysis of Experimental Data and Results

The experiment was conducted five independent replicates, with each control group subjected to three load levels (low, medium, high). Below are the original latency data (unit: ms) under moderate load for the hybrid control system:

number of replication	Task Registration Delay	Log Verification Delay	Control Response Delay	total delay
1	22.3	18.7	6.2	47.2
2	21.9	19.1	5.9	46.9
3	23.0	18.5	6.5	48.0
4	22.7	19.3	6.0	48.0
5	21.5	18.9	6.1	46.5
average	22.28	18.90	6.14	47.32

Average total latency in other control groups (medium load, 5 averages):

System Type	Average Total Delay (ms)	standard error
Single Agent	18.2	1.3
Partially Distributed	26.4	2.1
Centralized large-scale models	52.3	5.8
hybrid system	47.3	4.2

B.2 Statistical Analysis

b2.1 Variance Analysis (ANOVA) of Task Completion Rate

source of variation	quadratic sum (SS)	Freedom Degree (df)	Mean Square (MS)	F price	p price
Between groups	1892.6	3	630.9	28.4	<0.001
Within group (error)	355.8	16	22.2		
Total	2248.4	19			

Conclusion: $p < 0.001$, indicating statistically significant differences in task completion rates between different system types. Post-hoc Tukey tests revealed that the hybrid system significantly outperformed both single-agent systems ($p < 0.01$) and purely distributed systems ($p < 0.05$), while showing no significant difference compared to centralized large-model systems ($p = 0.23$).

b.2.2 Relevance and Regression Analysis

To evaluate the impact of blockchain latency on overall performance, linear regression was performed on data from 20 experiments conducted on the hybrid system.

Dependent variable: Task completion rate (%), Independent variable: Registration delay (ms)

regression equation :

$$\text{Completion Rate} = 98.2 - 0.153 \times \text{Registration Delay}$$

$$R^2 = 0.62, p = 0.008 \text{ (significant negative correlation)}$$

For every additional 1 ms of registration delay, the completion rate decreases by approximately 0.15 percentage points. Yet even under the worst-case scenario of 30 ms delay, the predicted completion rate remains above 93.5%.

B.3 Instructions for Handling Outlier Values

Abnormality detection criteria: The Z-score method is employed, with a threshold of $|Z| > 3$ defined as abnormal.

In the raw data, a single experiment conducted by the pure distributed system achieved only a 52% task completion rate ($Z = -3.2$) due to a sudden network disruption. Upon verification, this incident resulted from an unexpected physical network failure that exceeded normal system performance limits; thus it was excluded and replaced with the mean value from adjacent points using interpolation. After exclusion, the average completion rate of the pure distributed system was revised from 79.2% to 81.6%.

All final report data passed the Shapiro-Wilk normality test ($p > 0.05$) and the Levene's heteroscedasticity test ($p > 0.05$), meeting the prerequisites for parameter testing.

Appendix C: Glossary

term	English	explain
LLM	Large Language Model	Large Language Model
VLM	Vision-Language Model	Visual-Linguistic Model
MAS	Multi-Agent System	Multi-Agent Systems
Sim2Real	Simulation to Reality	The transition from simulation to reality
TPS	Transactions Per Second	Number of transactions per second
IPFS	InterPlanetary File System	Star File System
Hyperledger Fabric	—	Union Chain Framework
ROS2	Robot Operating System 2	Robot Operating System, 2nd Edition
CMPC	Cooperative Model Predictive Control	Cooperative Model Predictive Control
VLA	Vision-Language-Action	Visual-Linguistic-Action Model